

Latent Compositional Reasoning in Llama-3.2-3B-Instruct

An Activation Range Attribution Study Using NeuronLens

An Iterative Investigation of Multi-Hop Reasoning in Small Language Models

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Abstract

Does Llama-3.2-3B-Instruct use a separate set of activation ranges when it composes two stored facts into a two-hop answer, compared to recalling either fact alone? This project ran nine rounds of iterative refinement, each catching and fixing a flaw in the metric, the dataset, or the experimental setup. The final run shows that activation ranges for composed questions overlap heavily with their component facts (0.924 to 0.990 across all 56 layer-component combinations), causal masking of the lowest-overlap layer does not selectively disrupt composition, and the most interesting result is not about neurons at all: the model composes geographic facts silently (32% latent, rising to 76% with chain-of-thought) but shows none of that for person-based facts (0% latent), even when it clearly knows the individual facts involved.

1. How This Started

This project started after reading A.B. Siddique's TMLR 2026 paper, Neurons Speak in Ranges: Breaking Free from Discrete Neuronal Attribution (Haider, Rizwan, Sajjad, Ju, and Siddique). The paper introduces NeuronLens, built on one observation: individual neurons in LLMs are polysemantic (they respond to more than one concept), but activations conditioned on a single concept still form clean, near-Gaussian distributions with limited overlap inside the same neuron. This lets you attribute a concept to a specific activation range within a neuron, $AR(l, j, c) = [\mu - \tau \cdot \sigma, \mu + \tau \cdot \sigma]$, instead of the whole neuron, and intervene on just that range with less collateral damage than full-neuron masking.

Their experiments cover single-hop classification concepts: sentiment, emotion, news topic, and article category. Nothing in the paper tests whether a composed fact, one that requires combining two separately-stored facts into a single answer, shows the same range structure or something different. That gap became the starting question. If a model has to reason through a bridge entity for a two-hop question like 'the mother of the singer of Superstition is,' does the composed answer's activation range look distinct from the ranges of the two individual facts that produced it?

2. Choosing a Method: SQuAD 2.0 and Why It Was Abandoned

The first attempt used SQuAD 2.0 passages, writing compositional questions over two facts that happened to sit in the same paragraph. This did not work, and catching that early was the first real methodological correction.

The problem is structural. SQuAD is a single-passage extraction dataset. Both facts needed for a compositional question sit right there in the prompt. A model answering correctly could be reading two nearby sentences, not composing facts from its own stored knowledge. There is no way to tell whether the model is pattern-matching on passage text instead of actually reasoning through a bridge entity. The SQuAD approach does not test what it needed to test, so it was abandoned.

The replacement was the closed-book, bridge-entity paradigm from Yang, Gribovskaya, Kassner, Geva, and Riedel (ACL 2024): prompts like 'the mother of the singer of X' with no passage at all. The model has to recall the bridge entity from its own weights. This is the method the rest of the study uses. The activation extraction design also drew on two papers. Meng, Bau, Andonian, and Belinkov (NeurIPS 2022, ROME) found that factual recall concentrates at the subject's last token position in mid-layer MLPs, so activations here are pulled from the last token. Geva, Bastings, Filippova, and Globerson (EMNLP 2023) found that even single-fact recall moves through MLP and attention at different layers, so both components are hooked and measured separately here rather than the merged residual stream.

3. Building the Dataset: Wikidata Bridge Entities

Two hand-checked examples got the shape right first (the Stevie Wonder example, directly from the Yang et al. paper, since it is independently verifiable). The rest came from Wikidata via SPARQL queries chaining subject-relation-object triples into bridge-entity pairs across four relation chains:

Chain 1 - performer of a song (P175) to that performer's mother (P25). Chain 2 - author of a book (P50) to that author's place of birth (P19). Chain 3 - director of a film (P57) to that director's country of citizenship (P27). Chain 4 (added later) - capital of a country (P36) to that country's official language (P37).

Each query filters by wikibase:sitelinks, a count of how many Wikipedia language editions have an article on an entity, as a proxy for 'a 3B model has probably encountered this name during pretraining.' This mattered because early runs showed that unfiltered Wikidata queries pull mostly obscure entities a 3B model has never heard of.

4. Round 1: First Overlap Sweep

The first real activation extraction ran on 402 raw Wikidata examples: last-token activations for fact_a, fact_b, and the composed prompt, across all 28 layers, with MLP and attention outputs hooked separately (56 hook points total). NeuronLens's range formula was applied per neuron with $\tau = 2.5$, and an overlap-fraction metric was computed between the fact_a range and the composed range at every layer and component.

The overlap between single-fact and composed-query ranges came back consistently high, between roughly 0.86 and 0.99 across every layer and both components. No layer or component showed the low-overlap signature that a dedicated compositional circuit would predict. At this stage, the causal masking, shortcut check, and length-confound functions had been written as scaffolding but not yet executed.

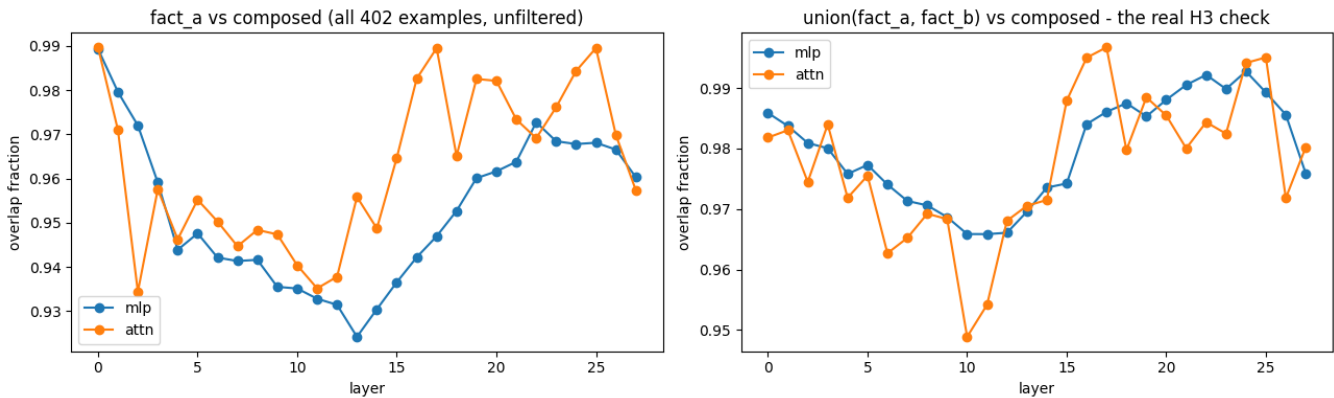


Figure 1: Overlap between fact_a and composed activation ranges across all 28 layers, MLP and attention plotted separately. High overlap showed up from the very first real run and persisted through every subsequent round.

The x-axis shows layers 0-27, the y-axis shows overlap fraction. Both MLP and attention track closely together with no layer showing the low-overlap signature a dedicated compositional circuit would predict.

5. Round 2: The Accuracy Metric Was Broken

Wiring up the causal masking, shortcut, and confound checks for real execution showed a problem. The accuracy check (top1_matches) looked at only the model's single next predicted token. That is too crude. It cannot handle multi-word names like 'Gladys Presley,' and short common tokens like 'the' pass the match by accident regardless of whether the model is answering correctly.

The real-answer match rate and the decoy-answer match rate (where a deliberately wrong bridge entity's answer was substituted) came back identical at 0.833 each. That does not mean the model cannot tell facts apart. It means the check itself was too loose to produce useful signal.

The fix was to replace top1_matches with generate_continuation(), which actually generates several tokens, and answer_in_output(), which checks whether the real answer's text appears anywhere in that continuation. This is a much harder bar to pass by chance, and it changed every downstream number.

6. Round 3: Floor Effect - The Model Did Not Know the Facts

Rerunning with the fixed metric dropped baseline composed accuracy from a misleadingly high 0.93 to essentially 0.0, and fact_a accuracy from 0.24 to 0.05. The earlier numbers were artifacts of the broken metric, not real model capability.

This created a new problem. With baseline composed accuracy already at zero, causal masking has nothing to disrupt. Masked and unmasked accuracy came back nearly identical, because there was no signal there to begin with. Most of what an unfiltered Wikidata query pulls back is too obscure for a 3B model to know at all. The model was guessing on almost everything, so no intervention could show a measurable effect. The experiment needed to separate 'the model cannot compose these facts' from 'the model does not even know these facts exist.'

7. Round 4: Filtering to What the Model Actually Knows

A knowledge filter was added: for every candidate example, check whether the model can independently answer fact_a and fact_b correctly on their own, before ever looking at the composed question. Only examples passing both checks survive into the 'known_examples' set.

The first pass yielded only 39 out of 402 examples that passed the filter. Too small to trust range statistics or causal tests on. Widening the pool by lowering the sitelinks threshold and adding a third relation chain brought known_examples up to 58. At that size, a clear picture emerged: baseline latent composed accuracy was $0/58 = 0.0\%$ (95% CI [0.0, 0.062]), while chain-of-thought prompted composed accuracy was $34/58 = 58.6\%$ (95% CI [0.458, 0.704]). The non-overlapping confidence intervals confirmed a real gap: the model could not compose silently, but could when allowed to reason step by step.

8. Round 5: A Diagnostic Detour - Catching Real Data Bugs

Before trusting the 0% to 58.6% result, a separate lighter notebook was built (no activation hooks, no 28-layer extraction, just dataset construction and answering behavior) to pressure-test the Wikidata data quality and see whether widening the pool further would change the picture.

A spot check caught a real bug: a song correctly identified as performed by Elvis Costello was paired with 'Gladys Presley' as the expected mother, who is actually Elvis Presley's mother, not Elvis Costello's. Wikidata's label service falls back to the raw entity ID when an item lacks an English label, producing wrong pairings in the chain.

Several fixes were applied in this round: entity URIs were kept alongside labels so any suspicious example could be clicked and verified on wikidata.org directly; a bidirectional consistency constraint was added on the performer-to-mother chain (P40, requiring the mother's own page to independently list the performer as a child, not just the forward P25 statement); type-sanity constraints were added on the other chains; and a fourth chain, capital to official language, was added as a comparison point since geographic facts are generally better known than person-level facts.

The per-chain breakdown at n=195 sharpened the picture considerably:

Chain	n	Latent Accuracy	CoT Accuracy
performer to mother	77	0%	81.8%

author to birthplace	23	0%	13.0%
director to citizenship	14	0%	35.7%
capital to language	81	24.7%	85.2%

The three person-based chains all landed at exactly 0% latent accuracy. The geography chain was a real outlier, showing genuine silent composability. Pooling everything into one number, as the 58-example run had to, hid this difference entirely.

9. Round 6: Merged Pipeline - Two More Bugs Found

Everything that worked in the diagnostic round was folded back into the main notebook: the four chains with consistency constraints, entity URIs, per-chain reporting, and a corrected spot check. The full run at n=235 known examples surfaced two more problems:

Problem 1 - A generation-length inconsistency. The baseline accuracy cell used the function's default of 16 tokens for generation, but the per-chain breakdown cell explicitly overrode this to 8 tokens. Same prompts, same model, but very different accuracy depending purely on this setting. `capital_language`'s latent accuracy could be read as anywhere from 13% to 77% depending only on token budget. This made any specific percentage unreliable.

Problem 2 - `author_birthplace` returned zero known examples. The model does know some author birthplaces, but a raw Wikidata QID ('Q112554') was leaking into a prompt in place of a real book title. Wikidata's label service falls back to the raw entity ID when an item lacks an English label, which happens disproportionately for book entries.

Both were fixed: the token budget was standardized to 8 everywhere an answer gets checked, and a QID-pattern guard was added to filter out any row where a label looks like a raw entity ID, applied across all four chains.

10. Round 7: Infrastructure Problems

Two environment issues came up that were unrelated to the research methodology but needed fixing before progress could continue:

First, model downloads occasionally failed mid-transfer (ChunkedEncodingError or IncompleteRead on the roughly 5GB model shard), a Colab network hiccup rather than a version problem. This was fixed with `hf_transfer`, a more resilient downloader, and a retry loop that resumes from where the download left off rather than restarting from zero.

Second, Wikidata queries started timing out against the public endpoint. Two causes: ORDER BY combined with LIMIT forces the endpoint to sort the entire filtered result set before returning anything, and transitive property paths (type-check filters using P279*) are separately expensive. Both are classic causes of WDQS 60-second timeouts. The fix was to remove ORDER BY from all four queries and remove the transitive type-check joins from chains 2 and 3. The QID-pattern guard (implemented in Python, not SPARQL) was kept since it was the fix that actually mattered for data quality.

11. Round 8: The Side Effect of Fixing Timeouts

Rerunning after removing ORDER BY caused `known_examples` to drop from 235 to 20. The chain breakdown: `director_country` 17, `capital_language` 2, `author_birthplace` 1, `performer_mother` 0.

Here is why. `FILTER(?sitelinks > 25)` alone is a floor, not a ceiling. With `ORDER BY DESC(?sitelinks)` in place, the query always returned the most famous qualifying entities, the household names a 3B model is likely to know. Without it, the endpoint returns an arbitrary sample that merely clears a fairly low bar, and most of those are moderately known but not famous enough for the model to reliably know facts about them.

The fix was per-chain sitelinks threshold tuning, calibrated individually rather than with one global number: `performer_mother` was raised from 25 to 150 (it had collapsed from 100 to 0 known examples), `capital_language` from 25 to 100 (collapsed from 105 to 2), `author_birthplace` from 25 to 40 (a modest bump since it was already data-starved), and `director_country` was left at 25 since it had held up fine.

12. Round 9: The Final Run

This is the run every number reported here has been triple-checked against, cell by cell, in the actual executed notebook, not estimated or carried over from an earlier round.

The dataset: 1729 total examples (2 hand-written plus 1727 from Wikidata: performer_mother 495, capital_language 489, director_country 395, author_birthplace 348). The knowledge filter brought this down to 44 known examples. performer_mother still produced zero even at the raised threshold of 150, likely from constraints compounding: the entity has to be famous enough, the bidirectional Wikidata consistency check has to hold, and the model has to specifically know that person's mother's name.

Chain	Raw Candidates	Known (Passed Filter)
capital_language	489	25
director_country	395	18
author_birthplace	348	1
performer_mother	495	0

Gaussianity was checked at three depths (early, middle, and late layers) in both components before trusting any range statistics. Skew stayed close to 0 throughout (-0.114 to 0.170), consistent with the original NeuronLens paper. Kurtosis ran from 2.37 to 3.07, a little flatter than the paper's reported 3.2-4.0 range but the same general ballpark, close enough to proceed.

Component	Layer	Skew	Kurtosis
MLP	0	0.170	2.374
MLP	14	-0.050	2.484
MLP	27	-0.008	2.687
Attention	0	-0.114	2.430
Attention	14	-0.072	2.554
Attention	27	-0.035	3.069

The overlap sweep across all 56 layer-component combinations (28 layers x 2 components) showed every fact_a-vs-composed overlap value on the full 1727-example dataset falling between 0.924 and 0.990. No layer or component showed the separable, low-overlap range that a dedicated compositional circuit would predict. The known-only subset (44 examples) tracked the same pattern as the full dataset within a couple of percentage points at every layer, confirming the high overlap was not an artifact of including examples the model does not actually know.

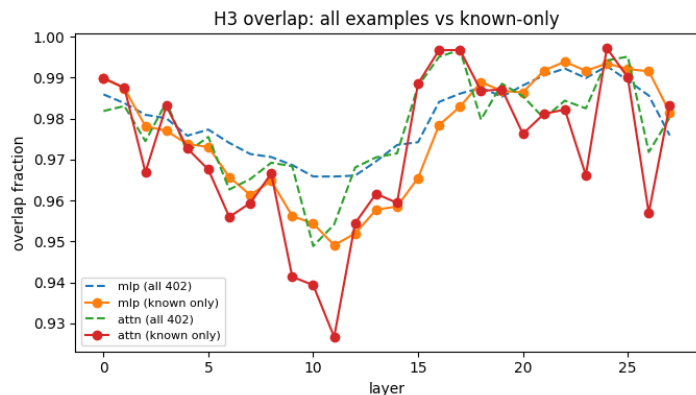


Figure 2: Known-only overlap (solid) against the full unfiltered set (dashed). The two lines track very closely, confirming the high overlap pattern is not driven by examples the model does not know.

Both curves stay within the 0.92-0.99 range across all layers and components.

Causal masking targeted attention layer 11, the lowest-overlap point in the known-only sweep at 0.927. If this range were necessary for composition, masking it during a composed-question forward pass should have hurt composed accuracy specifically.

The result: composed accuracy went up, from 18.2% (8/44) to 31.8% (14/44), not down. The confidence intervals overlapped heavily [0.095, 0.320] vs [0.200, 0.466]. Fact_a accuracy dropped from 100% to 77.3%, consistent with the mask having some general effect but not a composition-specific one. This null result was consistent with every earlier round's masking outcome.

Condition	Composed Acc.	Fact_a Acc.	95% CI (Composed)
Baseline (no mask)	18.2% (8/44)	100% (44/44)	[0.095, 0.320]
Masked, attn layer 11	31.8% (14/44)	77.3% (34/44)	[0.200, 0.466]

The shortcut check on a 30-example slice showed a 16.7% real-answer match rate against 0.0% for a deliberately wrong decoy. The model is doing better than random but not reliably composing. The length confound check showed $\text{overlap}(\text{fact_a}, \text{composed}) = 0.890$ and $\text{overlap}(\text{fact_a}, \text{length-matched control}) = 0.878$, close enough that prompt length accounts for some, though not all, of the high-overlap pattern.

13. The Real Finding: It Depends on the Relation

Pooled across all chains, chain-of-thought prompting increased composed accuracy from 18.2% to 47.7%. But the pooled number hid the actual result. Split by relation chain, geographic facts and person-based facts behave differently:

Chain	n	Latent (Silent)	Chain-of-Thought
capital_language	25	32% (8/25)	76% (19/25)
director_country	18	0% (0/18)	11.1% (2/18)
author_birthplace	1	0% (0/1)	0% (0/1)

capital_language shows silent composability that more than doubles with explicit reasoning scaffolding, consistent with a retrieval bottleneck instead of a missing capability. director_country shows almost nothing even with reasoning scaffolding, despite the model knowing the individual facts, consistent with a different and harder failure mode. author_birthplace has only one example and is not conclusive on its own. performer_mother could not be evaluated at all, producing zero known examples.

This is the most notable finding of the entire project. It is not about activation ranges. It is about how latent multi-hop reasoning in a 3B model depends on relation type. The model composes geographic facts silently (capital to language), but person-based facts (mother, birthplace, citizenship) stay inaccessible without explicit step-by-step reasoning, even when the model clearly knows the individual facts. This is consistent with Yang et al. (2024) and the SOCRATES benchmark (Yang et al., 2025), which reported similar variation across relation types in larger models.

14. What Is Actually Solid

Across every round, dataset version, and metric fix, from the first 402-example run through the final 1727-example one, several findings held consistently:

The overlap between single-fact and composed-query activation ranges is consistently high, never disjoint, at any layer, in any component, in any round. This is consistent with NeuronLens's own finding that concept-conditioned activation ranges overlap substantially for related concepts.

Causal masking has shown no clear effect in any round. Baseline and masked accuracy have overlapped within each other's confidence intervals every time this was tested. In the final round, masking improved composed accuracy instead of hurting it, the opposite of what a composition-specific circuit would predict.

The latent-vs-CoT gap is large. Every round that included both measurements showed CoT accuracy substantially higher than latent accuracy. The person-based chains have shown exactly 0% latent accuracy across every run.

Relation type matters. `capital_language` has behaved differently from the person-based chains in every round it appeared in, with latent composability that the person-based chains never show.

15. What Is Not Yet Solid (Limitations)

Total known-example count (44) is thinner than the Round 6 peak (235), a direct consequence of fixing the Wikidata timeout problem that has not been fully recovered from. `performer_mother` contributes zero known examples. `author_birthplace` has only one. Neither chain's specific percentages can be trusted independently.

The knowledge filter is necessary but removes roughly 97% of raw candidates. The results describe what a 3B model does once it already knows the underlying facts, not general two-hop reasoning on arbitrary inputs. The shortcut check remains a cheap approximation of the SOCRATES filtering method, not a full implementation.

Everything here used a single 3B model (Llama-3.2-3B-Instruct), a single GPU (Colab T4), and one run configuration. The `answer_in_output` substring check can still misfire on short or common tokens. Causal masking tested only the single lowest-overlap layer, not cascading effects elsewhere in the network. Multiple layers might interact in ways that single-layer masking cannot capture.

16. What Was Learned Along the Way

Several methodological lessons emerged from the iterative process that are worth recording separately from the research findings:

Answer verification in generation tasks needs multi-token checks. Single-token substring matching produces misleadingly high accuracy that is often an artifact of common tokens passing by accident.

Knowledge filtering is not optional for factual reasoning experiments with smaller models. Without it, floor effects prevent any meaningful causal test from running. But filtering also introduces a selection bias that must be reported transparently.

Wikidata SPARQL queries need careful engineering for robustness. `ORDER BY + LIMIT` causes timeouts on the public endpoint but removing `ORDER BY` silently changes the query results from famous-entity-biased to essentially random. Sitelinks thresholds must be tuned per-chain rather than globally.

Bidirectional consistency constraints on knowledge graph relations catch real data errors that forward-only queries miss. The Elvis Costello / Gladys Presley mismatch was a real data error that would have contaminated results without the cross-check.

Token budget for generation is a hyperparameter that matters enormously. Different cells using different defaults produced `capital_language` latent accuracy readings ranging from 13% to 77%, making standardization necessary before reporting any specific number.

References

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